

A Comprehensive Study of Edge Detection for Image Processing Applications

Ganesan P

Department of Electronics & Control Engineering
Faculty of Electrical and Electronics Engineering
Sathyabama University, Chennai, India
gganeshnathan@gmail.com

G.Sajiv

Department of Electronics & Telecommunication Engineering
Faculty of Electrical and Electronics Engineering
Sathyabama University, Chennai, India
sajiv.vijas@gmail.com

Abstract—In this paper, a comprehensive study of edge detection methods for image processing applications is carried out to analyze the various edge detectors and the latest trends in edge detection. An Edge in image processing can be described as discontinuities in intensity from one pixel to another. Edge detection is one of the most useful image enhancement techniques to improve the quality of the image analysis process. The principal objective of the edge detection is to identify and classify the discontinuities in an image. The edge detection in image processing considerably lessen the quantity of data (pixel) to represent an image and also filters out the futile information, while keeping the essential structural assets of an image. However, it is very difficult to perform edge detection in noisy images since it is uphill task to distinguish both the edges and noise in the image because both of them having high frequency components. In the past few decades, numbers of methods have been proposed for the detection of edges in color and intensity images. However, the edge detection is application (problem) oriented i.e., we can't apply a same algorithm for all types of images (applications). In this paper, an elaborative comparison of various edge detection methods for various image processing applications is performed.

Keywords—edge; edge detection; canny; sobel; prewitt; roberts wavelet transform

I. INTRODUCTION

The detection of edges in an image is the initial but important task in image analysis. This paper presents a comparative survey of various methods for the detection of edges. Edge detection plays an important role in image enhancement, analysis, compression, segmentation, object recognition and so on. [2]. Edge detection is one of the most useful image enhancement techniques to improve the quality of the image analysis process. The edge detection in image processing considerably lessen the quantity of data (pixel) to represent an image and also filters out the futile information, while keeping the essential structural assets of an image. However, it is very difficult to perform edge detection in noisy images since it is uphill task to distinguish both the edges and noise in the image because both of them having high frequency components.

The sketch out of the paper is as follows. An overview of literature survey on various edge detection methods is given the section 2. The edge detectors for image processing applications are explained in the section 3. The section 4 explains the experiments on the edge detection methods and finally the section 5 concludes the paper.

II. LITERATURE SURVEY

Gupta et. al., [1] had proposed edge detection based on the combined approach of particle swarm optimization and firefly optimization. Firefly optimization is modified using particle swarm optimization. The outcome of the proposed approach is compared with the existing traditional algorithms such as Prewitt, Canny, Sobel and Robert. Wenlong Fu et. al. [3] proposed a novel method of edge detection for the extraction of low level subjective features in natural using genetic programming. This automated technique of hunting pixels shuns the problem of blurring edges and also avoids noise influence. In GP edge detectors, linear and second-order filters are created from the pixels of high occurrences.

Wenshuo Gao et. al., [4] proposed a novel method for edge detection on noisy images. The combined approach of wavelet threshold and sobel edge detection to improve the standard sobel edge detector for the efficient detection edges. Large hardware unit and time complexity is the main limitations of the work. Kiani et.al., [5] explained an approach for the detection of edges based on the Shannon entropy in of remote sensing images. In the first step, thresholds of various parts (regions) of the image were computed in a piecewise manner. In the second step, using the proposed method, proper thresholds were extorted, and finally, Shannon entropy is applied to extract the boundary between these regions. The results were weighed against some standard edge detection algorithms

An edge detection technique based on the morphological gradient method and generalized type-2 fuzzy logic is explained in Melin et al.,[7]. In this work, the generalized type-2 fuzzy logic is implemented using the

theory of alpha planes for detection the detection of edges in an image. The approximation methods and heights are used for the defuzzification process. The suggested edge detection technique was investigated with synthetic images analyzed using Pratt measure to demonstrate the advantages of the proposed approach using generalized type-2 fuzzy logic.

Caixia Deng et.al.,[8] discussed an approach for the detection edges based on the sobel operator. Moreover, constructing the use of fusion method which combined an improved Sobel operator with wavelet transform to get better the accuracy of edge detection and obtain a fairly ideal edge detection effect. Shiling Sun et al., [9] proposed an edge detection algorithm based on the combined approach of particle swarm optimization (PSO) and fuzzy clustering method. The proposed approach overcomes the problem of local minimum and the initialization sensitivity of the traditional fuzzy clustering algorithm.

Hao et. al., [10] explained wavelet transform based image edge detection. This method is based on threshold denoising and wavelet transform to detect edges. Gautam et. al., [11] suggested Successive Otsu technique for the detection of edges. The proposed method is applied to partition an image histogram into many regions to present the threshold factor to standard Sobel Operator in order to improve its edge detection ability. Yanget et. al., [12] proposed dyadic wavelet transform based multi scale image edge detection. This work is based on the principles of wavelet transform modulus maxima theory and wavelet multi-scale features. Dyadic wavelet transform is applied to identify the wood defect (discontinuities) and the suppression of noisy information.

Di Jia et. al., [13] proposed an edge detection process of Gaussian block distance to acquire better edge information. At first, Gaussian weighted distance within sub-block regions is computed by computing the distance between the pixels. In the next step, edge detection method of block distance combining with summed area table is raised to reduce the complexity of the algorithm.

Anand et. al., [15] explained an edge detection approach for the medical images based on the mathematical morphology and Laplacian of Gaussian. This work merges the merits of both methods for the excellent detection of edges and good image contrast. This also presented the edge detection using various gradient based operators and mathematical morphology.

The Wavelet Transform has a dominant role in analyzing the 1D, 2D and 3D signals. In the field of image processing, the wavelet transform plays an important role in the detection of edges to choose the size of the image information that will be identified. A Wavelets transform split the lower and higher frequencies effortlessly, which is most important for the detection of edges. The scale of wavelet determines the size of the identified edges. In discrete wavelet transform based edge detection, wavelet

analysis is performed on the basis of horizontal and vertical function and edges are detected separately. Kumar et. al.,[16] proposed Daubachies wavelet transform based edge detection method for the decomposition of 2D images at three levels. At every level, the higher and lower frequency information of the image had splitted. The problem with the proposed approach is ghost edges which are solved using threshold.

III. EDGE DETECTION FOR IMAGE PROCESSING

In this section, the edge detection in images using edge detectors: Robert, sobel, Prewitt, Laplacian, canny and wavelet transform are explained.

Sobel Edge Detector

It has 2 3x3 convolution masks or kernels, Gx and Gy, as illustrated in fig. 1. These kernels can be used to find the orientation and magnitude of the gradient. The main application of sobel edge detection is to identify the edges in an image along the vertical and horizontal axis. So this edge detector tries to identify the edges in vertical and horizontal directions and then combined as a single metric.

$$\begin{bmatrix} -1 & 0 & 1 \\ 2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Fig. 1.3x3 convolution kernels or masks, Gx and Gy

$$|G| = \sqrt{G_x^2 + G_y^2}$$

Prewitt Edge Detector

The prewitt detector is very alike that of sobel operator. The characteristics of Prewitt edge detector is listed as

- Simplicity
- detection of vertical and horizontal edges and their orientations,
- sensitivity to noise, and
- inaccurate

The kernels can be used to find the orientation and magnitude of the gradient used in this prewitt detector is shown in fig. 2.

$$\begin{bmatrix} -1 & 0 & +1 \\ -1 & 0 & +1 \\ -1 & 0 & +1 \end{bmatrix} \quad \begin{bmatrix} +1 & +1 & +1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Fig. 2. Kernel of prewitt detector

Laplacian of Gaussian Edge Detector

Laplacian of Gaussian gradient edge detector is used to find out the regions of sudden intensity changes. In

this detector, the laplacian process is ended, then the image is applied to Gaussian filter to eliminate the unwanted noisy pixels. The laplacian gradient for a two dimensional image is given by

$$L(x, y) = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

The kernels can be used to find the orientation and magnitude of the gradient used in Laplacian of Gaussian gradient edge detector is shown in fig. 3.

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 8 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{bmatrix}$$

Fig. 3. Kernel of Laplacian of Gaussian edge detector

Roberts Edge Detector

In Roberts edge detector, one of the two kernels is shifted by 90° with respect to another. This edge detector performs a speedy 2D spatial gradient measurement on an image. The kernels can be used to find the orientation and magnitude of the gradient used in Roberts's edge detector is shown in fig. 4.

$$\begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix} \quad \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}$$

Fig. 4. Kernel of Roberts's edge detector

$$\Delta f = \text{grad}(f) = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \partial f / \partial x \\ \partial f / \partial y \end{bmatrix}$$

The magnitude of this edge detector is given by

$$M(x, y) = \sqrt{G_x^2 + G_y^2}$$

Canny Edge Detector

The canny edge detector can be used to identify a wide range of real edges in images. The detector eliminates the unwanted noise pixels by the process of smoothening edges in an image because noise pixels generate false edges. In this edge detection, the signal to noise ratio is improved as compared to other methods. This is the reason why this detector is extensively used for edge detection in image processing.

Wavelet based Edge Detection

Wavelet transform can be used for the detection of edges in intensity and color images. A two-dimensional discrete wavelet transform (2D-DWT) based edge detection in three various orientations such as diagonal, vertical and horizontal is illustrated in fig . In wavelet based edge

detection, the image is transformed into number of coefficients and then the decomposition of approximation coefficients at level j in four components.

The Wavelet Transform has a dominant role in analyzing the 1D, 2D and 3D signals. In the field of image processing, the wavelet transform plays an important role in the detection of edges to choose the size of the image information that will be identified. A Wavelets transform split the lower and higher frequencies effortlessly, which is most important for the detection of edges. The scale of wavelet determines the size of the identified edges. In discrete wavelet transform based edge detection, wavelet analysis is performed on the basis of horizontal and vertical function and edges are detected separately.

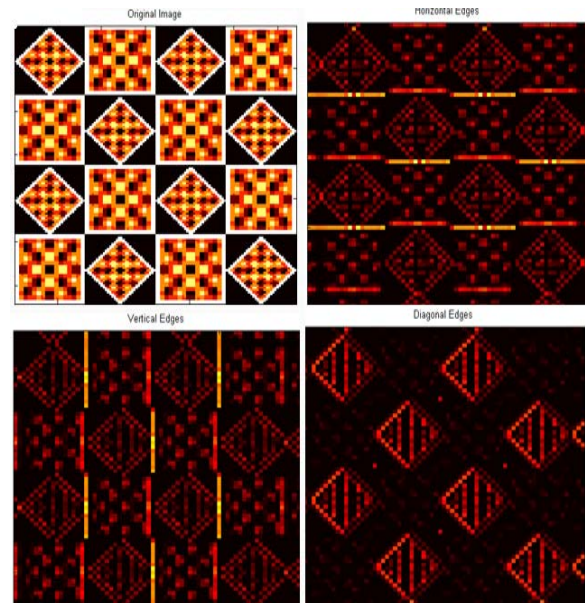


Fig. 5. Edge detection using discrete wavelet transform (a) input image (b) horizontal edge detection (c) vertical edge detection (d) diagonal edge detection (image courtesy :The Mathworks Inc)

TABLE I COMPARISON OF VARIOUS EDGE DETECTION TECHNIQUES

Edge Detector	Method	Advantages	Limitations
Roberts	Gradient Based	1. Easy and simple computation. 2. Edges are detected along with their orientation	1. These are more sensitive to noise. 2. Detection of edges is inaccurate 3. Less reliable
Sobel	Gradient Based	1. Easy and simple computation. 2. Edges are detected along with their orientation	1. These are more sensitive to noise. 2. Detection of edges is inaccurate 3. Less reliable
Prewitt	Gradient Based	1. Easy and simple computation. 2. Edges are detected along with their orientation	1. These are more sensitive to noise. 2. Detection of edges is inaccurate 3. Less reliable
Canny	Gaussian Based	1. Improved signal to noise ratio. 2. Suitable for noisy images i.e., more sensitive to noisy pixels 3. Accurate	1. Slow and Complex 2. False zero crossing.
LoG	Gradient Based	1. The detection of edges and their orientation is simple due to approximation of gradient magnitude is simple. 2. The characteristics are fixed in all directions. 3. Testing wide area around the pixel is possible.	1. Malfunctioning at the corners, curves and where the gray level intensity function varies 2.The magnitude of edges degrades as noise increases
DWT	Wavelet based	1.More accurate than other methods 2. Less computation	1.application oriented 2.complicated as compared to traditional methods

IV. EXPERIMENTS ON EDGE DETECTORS

The result of edge detection for different images is illustrated in fig 6 to fig 8.

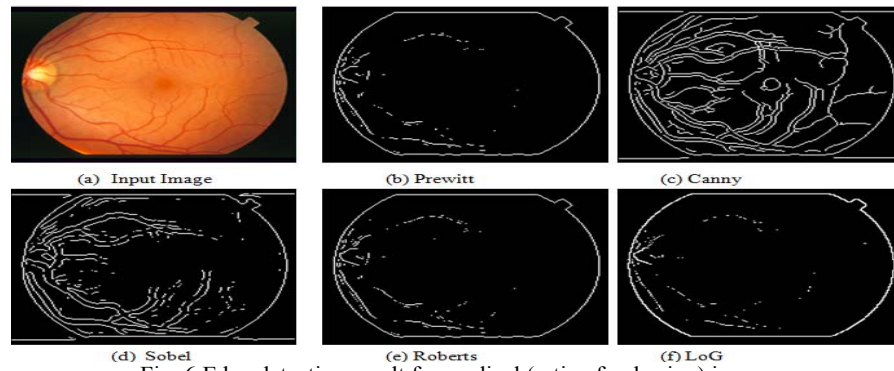


Fig. 6.Edge detection result for medical (retina fundus.jpg) image

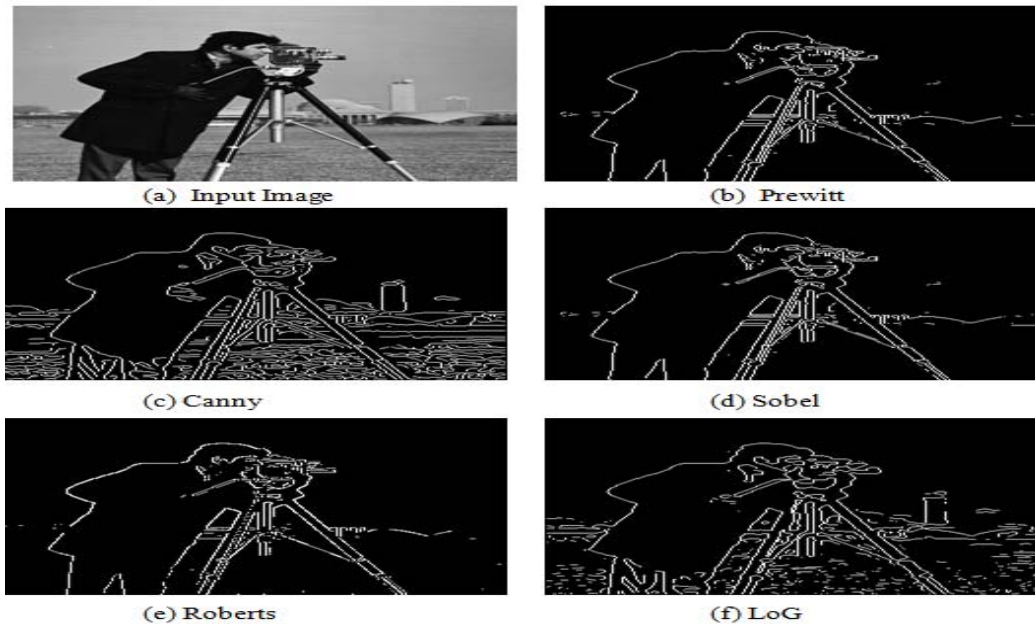


Fig. 7.Edge detection result for cameraman.tiff image

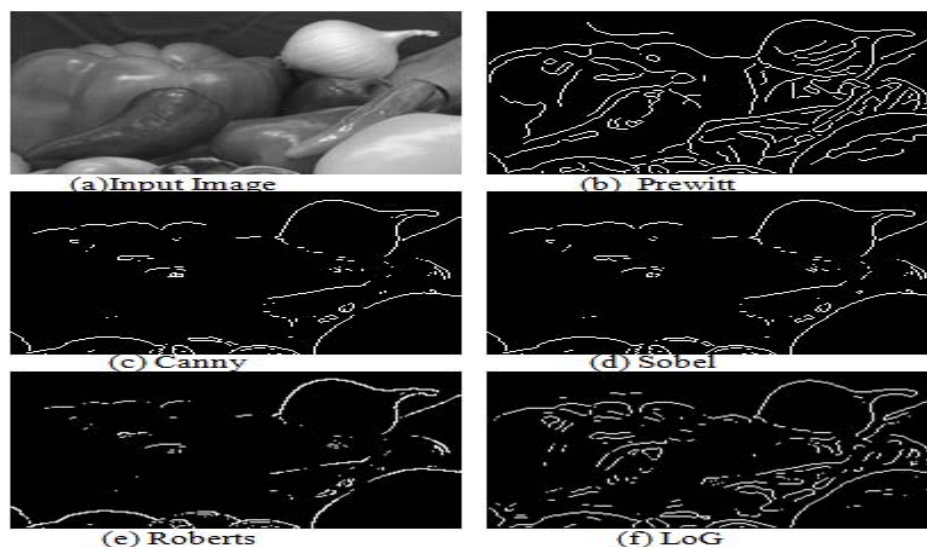


Fig. 8.Edge detection result for onion.png

V. CONCLUSION

This work analyzed and reviewed an extensive range of methods of edge detection for image processing applications. In the past few decades, numbers of methods have been proposed for the detection of edges in color and intensity images. However, the edge detection is application (problem) oriented i.e., we can't apply a same algorithm for all types of images. The performance evaluation of various edge detection methods is carried out for different images using Matlab. The merits and demerits of edge detection methods are tabulated. Among the standard edge detection methods, the canny edge detector produced very good results especially under noisy conditions. As far as accuracy concerned, Wavelets based methods are more accurate and outperformed other methods.

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